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# Safety data sheet



BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

Date / Revised: 02.01.2023 Version: 7.0
Date previous version: 11.03.2022 Previous version: 6.0
Date / First version: 13.03.2020
Product: **Sodium hypochlorite solution**

(ID no. 30042344/SDS_GEN_CH/EN)

Date of print 30.06.2025

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

# Sodium hypochlorite solution

UFI: 66JU-3FRD-A00T-1AUP

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Chemical
Recommended use: process chemical, oxidizing agents, Bleaching agents

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF Schweiz AG
Klybeckstrasse 161
4057 Basel, SWITZERLAND

Telephone: +41 0800 227722
E-mail address: PS-BCSCHWEIZ@basf.com

### 1.4. Emergency telephone number

Tox Info Suisse (STIZ): Tel. 145
International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

| | |
|---|---|
| Met. Corr. 1 | H290 May be corrosive to metals. |
| Skin Corr./Irrit. 1B | H314 Causes severe skin burns and eye damage. |
| Eye Dam./Irrit. 1 | H318 Causes serious eye damage. |
| Aquatic Acute 1 | H400 Very toxic to aquatic life. |
| Aquatic Chronic 2 | H411 Toxic to aquatic life with long lasting effects. |
| M-factor acute: 10 | |

For the classifications not written out in full in this section the full text can be found in section 16.

### 2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

| | |
|---|---|
| H290 | May be corrosive to metals. |
| H314 | Causes severe skin burns and eye damage. |
| H411 | Toxic to aquatic life with long lasting effects. |
| H400 | Very toxic to aquatic life. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P273 | Avoid release to the environment. |
| P260 | Do not breathe dust or mist. |
| P264 | Wash contaminated body parts thoroughly after handling. |
| P234 | Keep only in original packaging. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER or physician. |
| P303 + P361 + P353 | IF ON SKIN (or hair): Remove or Take off immediately all contaminated clothing. Rinse skin with water or shower. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |

| | |
|---|---|
| P301 + P330 + P331 | IF SWALLOWED: rinse mouth. Do NOT induce vomiting. |
| P391 | Collect spillage. |
| P390 | Absorb spillage to prevent material damage. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |
| P406 | Store in corrosive resistant container with a resistant inner liner. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):

EUH031: Contact with acids liberates toxic gas.

Hazard determining component(s) for labelling: Sodium hypochlorite

### 2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
The product is alkaline.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Not applicable

### 3.2. Mixtures

Chemical nature

Sodium hypochlorite (Content (W/W): >= 13 % - <= 16 %)NaOCl
CAS 7681-52-9 EINECS 231-668-3

dissolved in:Water

Regulatory relevant ingredients

Sodium hypochlorite

| | |
|---|---|
| Content (W/W): >= 13 % - <= 16 %<br>CAS Number: 7681-52-9<br>EC-Number: 231-668-3<br>INDEX-Number: 017-011-00-1 | Met. Corr. 1<br>Skin Corr./Irrit. 1B<br>Eye Dam./Irrit. 1<br>STOT SE 3 (irr. to respiratory syst.)<br>Aquatic Acute 1<br>Aquatic Chronic 1<br>M-factor acute: 10<br>M-factor chronic: 1<br>H290, H335, H314, H400, H410<br>EUH031<br><br>Specific concentration limit:<br>>= 5 % |

sodium hydroxide

| | |
|---|---|
| Content (W/W): >= 0,1 % - < 1 %<br>CAS Number: 1310-73-2<br>EC-Number: 215-185-5<br>INDEX-Number: 011-002-00-6 | Met. Corr. 1<br>Skin Corr./Irrit. 1A<br>Eye Dam./Irrit. 1<br>H290, H314<br><br>Specific concentration limit:<br>Skin Corr./Irrit. 1A: >= 5 %<br>Skin Corr./Irrit. 1B: 2 - < 5 %<br>Skin Corr./Irrit. 2: 0,5 - < 2 %<br>Eye Dam./Irrit. 2: 0,5 - < 2 % |

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

First aid personnel should pay attention to their own safety. Immediately remove contaminated clothing. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position).

If inhaled:
Inhale corticosteroid dose aerosol. Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200 - 300 ml water, do not induce vomiting, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Hazards: No hazard is expected under intended use and appropriate handling.

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote, administer corticosteroid dose aerosol to prevent pulmonary odema.

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
water spray

### 5.2. Special hazards arising from the substance or mixture

Advice: Vapors and/or decomposition products are irritant and/or toxic. Substance/product may act as an oxidizer.

Endangering substances: chlorine, sodium hydroxide
Advice: The substances/groups of substances mentioned can be released if the product is involved in a fire.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
If exposed to fire, keep containers cool by spraying with water. Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered.

## SECTION 6: Accidental Release Measures

Soiled textiles/cleaning rags made of natural fibres (e.g. of pure wool or of pure cotton) are capable of ignition and should not be used and/or must be desposed of in a safe manner.

### 6.1. Personal precautions, protective equipment and emergency procedures

Avoid inhalation. Avoid contact with the skin, eyes and clothing. Contact with natural fibres (e.g. of pure wool or of pure cotton) should be avoided because of possible ignition.

### 6.2. Environmental precautions

Do not empty into drains.

### 6.3. Methods and material for containment and cleaning up

For residues: Sweep/shovel up. Dispose of absorbed material in accordance with regulations.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

Pressure relief device necessary.

Protection against fire and explosion:
The substance/product is non-combustible.

### 7.2. Conditions for safe storage, including any incompatibilities

Segregate from acids.

Suitable materials for containers: High density polyethylene (HDPE), Bromobutyl rubber (BIIR) Vulcoferran 2208 (Steuler KCH), Bromobutyl rubber (BIIR) Vulcoferran 2208 T (Steuler KCH), Bromobutyl rubber (BIIR) HAW-W08 (HAW Linings), Bromobutyl rubber (BIIR) Chemoline 4, Chemoline RT (TIP TOP Elbe), Bromobutyl rubber (BIIR) Chemoline 8, Hypalon (TIP TOP Elbe), Bromobutyl rubber (BIIR) Vulcoferran 2206 (Steuler KHC), Bromobutyl rubber (BIIR) Vulcoferran 2209 T (Steuler KHC)
Unsuitable materials for containers: HAW-W12 (Hypalon, identical to Vulcoferran 2512, supplier HAW Linings GmbH), Compound based on HR004 / HR006 (supplier: Ragep)
Further information on storage conditions: Keep in a cool, well-ventilated place. Protect from the effects of light. Keep away from heat.

### 7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

1310-73-2: sodium hydroxide
STEL value 2 mg/m3 (MAC (CH)), Inhalable fraction
(MAC (CH)), Inhalable fraction
There is no reason to fear a risk of damage to the developing embryo or fetus when the occupational exposure limits or biological exposure limit (BEL) are adhered to.
TWA value 2 mg/m3 (MAC (CH)), Inhalable fraction

The substance mentioned develops if the regulation/notes for storage and handling are not observed.

7782-50-5: chlorine
STEL value 1,5 mg/m3 ; 0,5 ppm (MAC (CH))
TWA value 1,5 mg/m3 ; 0,5 ppm (MAC (CH))

Components with PNEC

7681-52-9: Sodium hypochlorite
freshwater: 0,00021 mg/l
marine water: 0,000042 mg/l
intermittent release: 0,00026 mg/l
sediment (freshwater):
Exposure of sediment is not expected
sediment (marine water):
Exposure of sediment is not expected
soil:
Exposure of soil is not expected
STP: 4,69 mg/l
oral (secondary poisoning): 11,1 mg/kg

1310-73-2: sodium hydroxide
freshwater:
According to EU risk assessment risks are negligible
marine water:
According to EU risk assessment risks are negligible
intermittent release:
According to EU risk assessment risks are negligible
sediment (freshwater):
According to EU risk assessment risks are negligible
sediment (marine water):
According to EU risk assessment risks are negligible
soil:
According to EU risk assessment risks are negligible

STP:
According to EU risk assessment risks are negligible

Components with DNEL

7681-52-9: Sodium hypochlorite
worker: Short-term exposure - systemic and local effects, Inhalation: 3,1 mg/m3
worker: Long-term exposure - systemic and local effects, Inhalation: 1,55 mg/m3
consumer: Long-term exposure - systemic and local effects, Inhalation: 1,55 mg/m3
consumer: Long-term exposure- systemic effects, oral: 0,26 mg/kg

1310-73-2: sodium hydroxide
worker: Long-term exposure - local effects, Inhalation: 1,0 mg/m3

## 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Self-contained breathing apparatus.

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
polyvinylchloride (PVC) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness
butyl rubber (butyl) - 0.7 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Protective suit, chemical-protection suit (f.e. according to EN 14605)

General safety and hygiene measures

Take off immediately all contaminated clothing.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | liquid | |
| Form: | solution, liquid | |
| Colour: | yellow to green | |
| Odour: | pungent odour<br>of chlorine | |
| Odour threshold: | Not determined due to potential health hazard by inhalation. | |
| Melting point: | -30 - -20 °C | (other) |
| Boiling point: | 100 °C<br>(1.013 mbar)<br>Information applies to the solvent.<br>The substance / product decomposes. | |
| Lower explosion limit: | For liquids not relevant for classification and labelling. | |
| Upper explosion limit: | For liquids not relevant for classification and labelling. | |
| Flash point: | not applicable | |
| Auto-ignition temperature: | not applicable | |
| Thermal decomposition: | Decomposes on heating. | |
| pH value: | 12<br>(160 g/l) | (OECD Guideline 122) |
| Viscosity, dynamic: | 3 - 4 mPa.s<br>(20 °C) | (OECD 114) |
| Solubility in water: | readily soluble<br>(15 °C) | |
| Partitioning coefficient n-octanol/water (log Kow): | The value has not been determined because the substance is inorganic. | |
| Vapour pressure: | 20 mbar<br>(20 °C) | (measured) |
| Density: | 1,24 - 1,26 g/cm3<br>(20 °C) | |

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

### 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

**Other safety characteristics**

Miscibility with water: completely (e.g. >=90%)

Evaporation rate: Value can be approximated from Henry's Law Constant or vapor pressure.

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

### 10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions

Reacts with acids. Exothermic reaction.

### 10.4. Conditions to avoid

Avoid extreme temperatures.

### 10.5. Incompatible materials

Substances to avoid:
acids

### 10.6. Hazardous decomposition products

Hazardous decomposition products:
chlorine

## SECTION 11: Toxicological Information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Experimental/calculated data:
LD50 rat (oral): > 5.000 mg/kg
The statement for acute oral toxicity was derived from products of similar composition. Literature data.

LD50 rabbit (dermal): > 5.000 mg/kg
The statement for acute dermal toxicity was derived from products of similar composition. Literature data.

*Information on: Sodium hypochlorite*
*Assessment of acute toxicity:*
*The toxicity of the product is based on its corrosivity.*
---

Irritation

Assessment of irritating effects:
Corrosive! Damages skin and eyes. Risk of serious damage to eyes.

Experimental/calculated data:
Skin corrosion/irritation
rabbit: Corrosive.
The product has not been tested. The statement has been derived from the properties of the individual components.

Serious eye damage/irritation
rabbit: irreversible damage
The product has not been tested. The statement has been derived from the properties of the individual components.

Respiratory/Skin sensitization

*Information on: Sodium hypochlorite*
*Assessment of sensitization:*
*Skin sensitizing effects were not observed in animal studies.*
---

Germ cell mutagenicity

*Information on: Sodium hypochlorite*
*Assessment of mutagenicity:*
*The substance was mutagenic in various test systems with microorganisms and cell cultures; however, these results could not be confirmed in tests with mammals.*
---

Carcinogenicity

*Information on: Sodium hypochlorite*
*Assessment of carcinogenicity:*
*In long-term studies in rats and mice in which the substance was given by drinking-water, a carcinogenic effect was not observed.*
---

Reproductive toxicity

*Information on: Sodium hypochlorite*
*Assessment of reproduction toxicity:*
*No data available. The chemical structure does not suggest a specific alert for such an effect.*
---

Specific target organ toxicity (single exposure)

No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

No data available.

Aspiration hazard

Study does not need to be conducted.

Interactive effects

No data available.

## 11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting

properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
Very toxic (acute effect) to aquatic organisms. Very toxic to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) 0,01 - 0,1 mg/l, Fish
The ecological data given are those of the active ingredient.

Aquatic invertebrates:
EC50 (48 h) 0,01 - 0,1 mg/l, daphnia
The ecological data given are those of the active ingredient.

Microorganisms/Effect on activated sludge:
Toxic limit concentration 0,375 mg/l, activated sludge
Literature data.

*Information on:Sodium hypochlorite*
*Assessment of aquatic toxicity:*
*Very toxic (acute effect) to aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.*
---

*Information on:Sodium hypochlorite*
*Aquatic plants:*
*EC50 (168 h) approx. 0,023 mg/l (other), unspecified algae (other, Flow through.)*
*Literature data.*

*No observed effect concentration (168 h) 0,0021 mg/l (other), unspecified algae (other, Flow through.)*
*Literature data.*
---

### 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Inorganic product which cannot be eliminated from water by biological purification processes. The product can be degraded abiotically, e.g. chemical or photolytic processes.

Information on Stability in Water (Hydrolysis):
$t_{1/2}$ 2 h

The action of light on the surface strat in water will induce decomposition.

### 12.3. Bioaccumulative potential

Bioaccumulation potential:
Accumulation in organisms is not to be expected.

### 12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: No data available.

### 12.5. Results of PBT and vPvB assessment

The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

### 12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

### 12.7. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### 12.8. Additional information

Adsorbable organically-bound halogen (AOX):
The Substance/product may have a halogenizing effect and therefore contribute to the OBH.

Other ecotoxicological advice:
Because of harmful effects on water organisms should not be introduced into drains. Do not discharge product into the environment without control. Do not discharge substance/product into sewer system. The substance/ product may be toxic to aquatic organisms in effluent treatment plants or surface waters by splitting of reactive substance groups. Very toxic (acute effect) to aquatic organisms.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Reduce with sodium sulphite, sodium pyrosulphite or sodium thiosulphate.

Contaminated packaging:
Transport containers should be completely emptied and returned.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN1791 |
| UN proper shipping name: | HYPOCHLORITE SOLUTION |
| Transport hazard class(es): | 8, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | Tunnel code: E |

RID

| | |
|---|---|
| UN number or ID number: | UN1791 |
| UN proper shipping name: | HYPOCHLORITE SOLUTION |
| Transport hazard class(es): | 8, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN1791 |
| UN proper shipping name: | HYPOCHLORITE SOLUTION |
| Transport hazard class(es): | 8, EHSM |
| Packing group: | II |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 1791 |
| UN proper shipping name: | HYPOCHLORITE SOLUTION |
| Transport hazard class(es): | 8, EHSM |
| Packing group: | II |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-B |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 1791 |
| UN proper shipping name: | HYPOCHLORITE SOLUTION |
| Transport hazard class(es): | 8 |
| Packing group: | II |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards

See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (2) significantly water polluting.

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

Annex 1 of the Ordinance on Protection against Major Accidents contains criteria for determining maximum levels based on toxicity, flammability, explosiveness, and eco-toxicity.

The following Swiss regulations must be observed for the professional use:

- Article 4 Paragraph 4 of the Ordinance on the Protection of young workers (SR 822.115) and Article 1 letter f of the EAER Ordinance on Dangerous Work for Young People (SR 822.115.2): Young people in an initial professional training can only work with this product (this substance / preparation) if this is foreseen in the respective education ordinance to achieve their education goal, the requirements of the education plan are fulfilled and the applicable age restrictions are observed. Young people who do not complete any initial professional training are not allowed to work with this product (this substance / this preparation). Employees of both genders up to the age of 18 are considered as young people.

### 15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

## SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Eye Dam./Irrit. 1
Skin Corr./Irrit. 1B
Met. Corr. 1
Aquatic Acute 1
Aquatic Chronic 2

M-factor acute: 10

| | |
|---|---|
| Met. Corr. | Corrosive to metals |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| STOT SE | Specific target organ toxicity — single exposure |
| H290 | May be corrosive to metals. |
| H314 | Causes severe skin burns and eye damage. |
| H411 | Toxic to aquatic life with long lasting effects. |
| H400 | Very toxic to aquatic life. |
| H335 | May cause respiratory irritation. |
| H410 | Very toxic to aquatic life with long lasting effects. |
| EUH031 | Contact with acids liberates toxic gas. |

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.